

# Sindhu Motapothula

## Data Engineer

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### PROFESSIONAL SUMMARY

Data Engineer with 6+ years of experience designing, building, and optimizing **data pipelines, analytics platforms, and machine learning workflows** across **Azure, AWS, GCP, and Snowflake** environments. Proven expertise in **big data processing (Spark, PySpark, Kafka, Flink)**, **ETL/ELT development**, and **cloud-native data architecture** for large-scale financial, retail, and healthcare systems. Skilled in **SQL, Python, and dbt** for transforming raw data into actionable insights, improving data quality, and enabling data-driven decision-making. Strong background in **data lakehouse design (Delta, Iceberg, Hudi)**, **orchestration (Airflow, Jenkins, GitHub Actions)**, and **CI/CD automation**. Adept at collaborating with cross-functional and **ML teams** to operationalize models in **SageMaker, Vertex AI, and Azure ML**, driving measurable gains in analytics efficiency, reliability, and compliance.

### TECHNICAL SKILLS

**Programming & Scripting:** Python, SQL, Scala, PL/SQL, Shell Scripting (Unix)

**Big Data & Distributed Computing:** Apache Spark, Hadoop, Databricks, Microsoft Fabric (OneLake, Lakehouse, Data Warehouse)

**Data Engineering & ETL/ELT Tools:** dbt Core, Apache Airflow, AWS Glue, Azure Data Factory, SSIS, Dataflow Gen2

**Cloud Platforms:** AWS (EC2, S3, Glue, Redshift, RDS, DynamoDB, Lambda, Kinesis), Microsoft Azure (Databricks, Synapse Analytics, Data Lake, Azure ML, ADF, Purview), GCP (BigQuery, Pub/Sub, Dataflow, Vertex AI)

**Databases & Data Warehousing:** Snowflake, SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, Bigtable, Azure Synapse Analytics

**Streaming & Real-Time Data Processing:** Apache Kafka, Amazon Kinesis, Azure Event Hubs, Apache Flink, Spark Streaming

**Machine Learning & AI:** Scikit-learn, XGBoost, TensorFlow, Azure Machine Learning, Amazon SageMaker, Pandas, NumPy, Feature Engineering, Model Deployment, Hyperparameter Tuning, Machine Learning Pipelines

**DevOps, CI/CD & Infrastructure:** Git, GitHub Actions, Jenkins, Azure DevOps, Docker, Kubernetes, Terraform

**Data Modeling, Governance & Lakehouse:** Delta Lake, Apache Iceberg, Parquet, ORC, Data Cataloging, Metadata Management

**Data Visualization & Business Intelligence:** Power BI, Tableau, Excel, Matplotlib, SSRS

**Methodologies & Practices:** Agile, Scrum, Software Development Life Cycle (SDLC), MLOps, DataOps

### WORK EXPERIENCE

Reprise Financial | TX | *Data Engineer*

July 2025 – Present

- Architected and executed the migration of **3+ TB of core financial data** from a legacy **Azure Synapse/ADF** environment to **Microsoft Fabric (OneLake)**, completing the project **10% ahead of schedule**.
- Implemented robust **ETL/ELT pipelines** using **Apache Spark (PySpark)** and **Dataflow Gen2**, transforming raw financial data into **curated medallion layers**, cutting manual data preparation time by **25%**.
- Developed and governed a **dbt Core** transformation layer with **SQL models** to standardize business logic, elevating **data consistency** and **definition alignment** across analytics platforms by **20%**.
- Authored **Python-based machine learning models** with **scikit-learn** and **XGBoost** to predict customer loan default risk, boosting model accuracy by **12%** and enabling more accurate **data-driven lending decisions**.
- Automated and orchestrated **data and machine learning pipelines** via **Apache Airflow** on **Azure**, integrating **CI/CD pipelines** with **GitHub Actions** to improve deployment reliability and reduce manual oversight by **40%**.

Netflix | CA | *Data Engineer*

June 2023 – June 2025

- Designed high-throughput **data pipelines** on **AWS** using **EMR (Apache Spark)**, **AWS Glue**, and **Amazon Kinesis**, delivering a **25%** improvement in data ingestion for content and user activity.
- Built **real-time event processing systems** using **Apache Kafka (MSK)** and **AWS Lambda**, enabling near-instant analytics for **A/B testing, content recommendations, and user engagement tracking**.
- Optimized **data transformation scripts** in **PySpark** and **Python (Pandas)** with **Apache Iceberg** and **Delta Lake table** formats, improving analytical query performance by **20%**.
- Orchestrated complex data workflows using **Apache Airflow** and **Meson (Netflix's orchestration platform)**, improving **SLA adherence** by **15%** and minimizing manual intervention.
- Streamlined **CI/CD pipelines** for data workflows using **Jenkins** and **Spinnaker**, reducing deployment failures by **30%** and introducing **seamless rollback** capabilities for production environments.
- Enhanced **Amazon S3 data lakes** through **partitioning, Parquet/ORC compression**, and **AWS Athena** optimization, reducing **query costs** by **18%** and improving **low-latency data access**.
- Collaborated with **Machine Learning teams** to **operationalize feature pipelines** in **Amazon SageMaker** and **Feast (feature store)**, ensuring reliable delivery of **high-quality ML features** for personalized recommendation models.

- Constructed and maintained **data pipelines** on **Google Cloud Platform (GCP)** using **BigQuery**, **Cloud Dataflow (Apache Beam)**, and **Pub/Sub**, driving large-scale **real-time sales** and **inventory analytics**.
- Developed **real-time streaming pipelines** using **Apache Kafka** and **Apache Flink**, enabling instant analytics for **demand forecasting**, **dynamic pricing**, and **fraud detection** across digital and in-store systems.
- Automated **ETL workflows** in **Apache Spark (Databricks)** with **SQL-based transformations**, improving data processing efficiency by **22%** for **supply chain**, **pricing**, and **customer analytics**.
- Designed scalable **data models** in **Bigtable** for high-speed transactional processing and **Snowflake** for enterprise-level reporting, optimizing **data retrieval** and **storage performance**.
- Partnered with **merchandising** and **supply chain teams** to integrate pipelines with **Vertex AI** for **machine learning–driven inventory optimization** and **predictive demand modeling**.

- Refined **data warehousing** in **Synapse Analytics** and **PostgreSQL**, enhancing query performance by **25%** for large-scale datasets while guaranteeing regulatory auditability.
- Built scalable **data processing systems** in **PySpark (Spark SQL/DataFrames)** on **Azure Databricks**, reducing processing time for **omics data (genomics, proteomics)** by **30%** in a fault-tolerant distributed environment.
- Developed and automated **data integration workflows** using **Azure Data Factory** and **Python**, processing **clinical trial datasets** in compliance with **HIPAA** and **GxP** standards, boosting integration efficiency by **20%**.
- Deployed **cloud-native analytics** via **Azure Data Lake Storage**, **Synapse Serverless**, and **Azure Purview**, incorporating **Parquet partitioning** and **metadata governance** to accelerate research insights.
- Containerized **data applications** with **Docker** and **Azure Container Instances (ACI)**, streamlining deployment of **ML models (Azure ML Service)** and **bioinformatics pipelines**, improving **reproducibility across research environments**.
- Managed validation and deployment workflows using **Azure DevOps Pipelines**, reducing pipeline errors by **30%** and ensuring **FDA 21 CFR Part 11 compliance**.

## **EDUCATION**

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**Master's in Healthcare Informatics,**

Saint Leo University, Tampa, Florida, USA.